

File Storage

CRM/ERP Platform Documentation

File Storage

Files are stored on the local filesystem under `backend/uploads/`.

Upload Endpoints

Endpoint	Purpose	Max Size
<code>POST /files/upload</code>	Record attachments	10 MB
<code>POST /gallery/upload</code>	Gallery images	10 MB
<code>POST /portal/documents</code>	Portal user documents	10 MB
<code>POST /public/forms/:token/extract-document</code>	OCR document	10 MB

All upload endpoints accept `multipart/form-data`.

Storage Structure

```
backend/uploads/  
└─ records/  
   └─ {orgId}/  
      └─ {timestamp}-{random}-{originalname}
```

The `File` model stores:

- `url`: The path used to access the file (e.g., `/api/v1/files/serve/{orgId}/{filename}`)

- `path` : Absolute path on disk
 - `originalName` : Original filename from the upload
 - `mimeType` , `size`
-

Serving Files

Files are served via:

```
GET /api/v1/files/serve/:orgId/:filename
GET /api/v1/gallery/serve/:orgId/:filename
```

These endpoints require **no authentication** — the URL itself is the access control (long random token in the filename). Keep file URLs private.

Multer Configuration

The backend uses `multer` with `diskStorage` :

```
destination: (req, file, cb) => {
  cb(null, `uploads/records/${orgId}/`)
},
filename: (req, file, cb) => {
  const uniqueSuffix = `${Date.now()}-${Math.random().toString(36).slice(2)}`;
  cb(null, `${uniqueSuffix}-${file.originalname}`)
}
```

Production Considerations

Persistence

If running in a container or on a PaaS (Railway), the `uploads/` directory is **ephemeral** by default. Set up a persistent volume:

```
# docker-compose.yml

volumes:
  - ./backend/uploads:/app/uploads
```

On Railway, attach a persistent volume to the `/app/uploads` path.

Backup

Include `backend/uploads/` in your backup routine. See [Backup-Recovery.md](#).

Cloud Storage (Future)

The current implementation uses local disk. For production scalability, swap `multer diskStorage` for an S3-compatible provider (AWS S3, Cloudflare R2, MinIO). This requires code changes in the files service — the `url` field would point to the CDN instead of the local serve endpoint.

Gallery

The gallery (`/gallery`) is a media library for the organization. Gallery files are stored under `uploads/gallery/{orgId}/`.

Gallery features:

- Category organization
 - Tag-based search
 - Download count tracking
 - Archive/unarchive
 - Used as cover images for Publications
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Portal Documents

Portal users can upload documents that are stored as `PortalDocument` records. These are linked to the portal user and optionally to a CRM record.

Admins can view portal documents via `GET /portal/padmin/documents`.

File Size Limits

The 10 MB limit is set in `backend/src/main.ts` :

```
app.use(express.json({ limit: '10mb' }));  
app.use(express.urlencoded({ extended: true, limit: '10mb' }));
```

To increase it, update both the Express limits and the Multer config in the relevant service.